Protecting Images from Diffusion-Based Geolocalization
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Abstract

Recent advances in generative models, whether through diffusion models or Vision Language Models, have demonstrated remarkable capabilities in inferring the geolocation of an image from its pixel content alone. This raises serious privacy concerns, as seemingly innocuous images shared online can now unknowingly reveal sensitive geographic information. Our paper addresses this issue in two connected directions. We first study the ability of state-of-the-art geographic models to detect whether the GPS metadata of an image has been tampered with. We then study two complementary methods to protect images from such models, by crafting imperceptible adversarial perturbations designed to mislead their geolocation predictions. We present Diffusion Trajectory Deviation, our novel adversarial attack that allows for efficiently protecting an image's location across two benchmark models and datasets.

1. Introduction and related work

Recent advances in generative models have enabled generalist or specialized models to infer geographic positions from images, sparking increased confidentiality concerns. Following a span of literature preoccupied with crafting methods to preserve geo-privacy, we investigate the capabilities of recent geographic diffusion and flow matching models [4]. We demonstrate that these specialized models can detect when an image's reported GPS location is false with an accuracy ranging up to 75% depending on the magnitude of the *lie*, i.e. on how far the false location is from the real one. This means that manipulating metadata to protect geo-privacy is insufficient.

We therefore develop a geo-privacy preserving framework, relying on adversarial attacks on images. We carefully train perturbations of images in order to prevent geographic diffusion models from predicting their location. Such attacks are difficult to craft, because training them directly by backpropagating through the diffusion process is prohibitively costly. We hence evaluate encoder attacks, and develop a novel method, **Diffusion Trajectory Deviation**. This novel approach targets the geographic diffusion process directly while preserving low training costs, and yields consistent results across evaluation datasets and models - despite encoder attacks also displaying some success.

Our contributions are threefold:

1. We highlight the strength of state-of-the-art geographic diffusion models in GPS metadata manipulation detection, which raises confidentiality concerns.
2. We propose a novel adversarial attack against diffusion models, Diffusion Trajectory Deviation, which achieves stealthy, efficient and effective perturbation of geographic diffusion models
3. We evaluate existing attacks and our new framework on two datasets, and highlight the usefulness of adversarial geo-privacy preservation.

Global visual geolocation. Global visual geolocation is a task aimed at predicting an image's localization using only its pixel information (no metadata) [9]. Recent advances have focused on treating visual geolocation as a generative task, rather than regression or classification, and have achieved substantial improvements in prediction accuracy [4, 22]. Dufour et al. [4] especially achieved state of the art performance by training geographic diffusion and flow matching models conditioned on encoded images via *DinoV2* or *StreetClip*. Visual geolocation has also risen as an emerging capability of Vision Language Models (VLM) [14, 24], fuelling efforts to fine tune multimodal foundational models for visual geolocation [22].

Adversarial attacks on diffusion models. Recent advances in image generation, in particular diffusion models [10, 19], have sparked concerns regarding the privacy of training data as well as potential malicious utilization of such models to generate harmful content. A typical example could include generating *deepfakes* of a politician based on her recent publications.

Adversarial attacks on diffusion models were introduced

to address this concern. Building on the existing literature on adversarial attacks (cite), Salman et al. [18] propose attacks that perturb images to prevent AI-powered editing. They also highlight the challenge of this task, due to the iterative nature of diffusion which makes directly training an attack by backpropagating through the diffusion inference process prohibitively costly and hard due to vanishing gradients. Kang et al. [11] propose *DiffAttack*, which optimizes a deviated-reconstruction loss at intermediate diffusion steps in order to avoid vanishing gradients, and backpropagate only through a few diffusion steps to avoid memory errors - a method used by Salman et al. [18] to craft a direct attack on diffusion models.

Geographic privacy preserving adversarial attacks. Geographic privacy has received specific attention in light of the aforementioned privacy concerns [16]. Existing general attacks against VLMs perturb image features to align them with a target image [6, 12, 23]. This can be used to modify an image's predicted location, although it comes at the expense of distorting all image semantics, not just geo-revealing ones. Liu et al. [13] hence introduce *GeoShield*, an adversarial framework to protect the geo-privacy of images against VLMs by targeting geo-revealing regions and features of images. Their works relies on untargeted attacks, and succesfully protects geo-privacy while preserving other image semantics.

2. Method

In this section, we present our methods to evaluate an image's vulnerability to localization using Dufour et al. [4]'s diffusion and flow matching models. We then present two adversarial methods to protect images from such localization.

Notation. Following Dufour et al. [4], we denote c an image to attack located at x_0 (denoted as x_0^c when the link to c must be specified). This position can be inferred through a Diffusion Denoising Implicit Model (DDIM) process relying on the denoising network ψ. It results in predicted coordinates$\hat{x}_t$ at timestep t . We consider a perturbed image $c + \delta$, and denote its predicted position by $\hat{x}_0^{c+\delta}$.

2.1. Evaluating images' vulnerability to localization

In this section, we explore how diffusion models can help determine if the geolocation attached to an image is real or fake. We mainly lean our work on the work of Dufour et al. [4] and more precisely on their flow-matching approach that allows us to compute likelihood $p(y|c)$ for any image c and any location y.

Threshold-based model. Our first approach is to achieve our binary classification using a threshold τ whose goal is to separate likely pairs (y, c) from unlikely ones. We try to empirically seek the best threshold.

Localizability based model. We then tried to refine this idea by using the localizability of an image.

We use the following localizability metric from Dufour et al. [4]:

$$Localizability(c) = \int_{\mathcal{S}^2} p(y|c) log_2 p(y|c) dy, \quad (1)$$

for $p(y|c)$ the distribution of c's localization on the sphere $\mathcal{S}^2$. This allows to evaluate spatial ambiguity, and hence potentially the need for geo-privacy preserving perturbations (the more localizable an image is, the less is preserves geo-privacy). It can be computed by using a Monte-Carlo method relying on the fact that $y_i \sim p(y|c)$.

We therefore have $\frac{1}{N}\sum_{i=1}^{N} \log_2 p(y_i \mid c)$ that is a non-biased estimator of the localizability and which is directly comparable to the log-likelihood of a pair (y, c) (as a mean of log-likelihood).

We have therefore crafted a binary classifier comparing $\log_2 p(y \mid c) - \frac{1}{N}\sum_{i=1}^{N} \log_2 p(y_i \mid c)$ with a given Δ threshold.

We set $N = 10$, which is a lot lower than the actual $N = 1024$ used originally for runtime reasons. We then try to empirically seek for the best Δ threshold.

2.2. Attacking geographic diffusion

In this section, we present two adversarial attacks to target geographic diffusion conditioned on an image. Here, we focus on the diffusion version of the models presented by Dufour et al. [4] for simplicity, although our approach could be extended to flow matching.

The first method, which we call *encoder attack*, is directly derived from Salman et al. [18] and is unspecific to (geographic) diffusion. We then present our new attack, *Diffusion Trajectory Deviation* (DTD) which directly targets the diffusion process itself, while avoiding backpropagating through the sampling process.

Model to attack. The geographic diffusion model to attack is directly taken from Dufour et al. [4]. It consists of a denoising neural network ψ trained to minimize the L^2 distance to random noise ϵ added to the image c's GPS coordinates x_0. The image-coordinate pairs are distributed along Ω which is dataset dependent. Following the literature [], we set $(\epsilon, t) \sim \mathcal{N}(0, I) \otimes \mathcal{U}[0, 1]$.

Hence, ψ minimizes the following training loss:

$$\mathcal{L} = \mathbb{E}_{x_0, c, \epsilon, t}[||\psi(x_t|c) - \epsilon||^2] \quad (2)$$

The neural network takes as input an image c and a noised coordinate x_t. It can be decomposed in two main components: an encoder ε which maps an image into a lower dimensional latent space, and a sub network ϕ which

predicts the noise from x_t and conditioned on the encoded image via adaptive layer normalization [21]:

$$\psi(x_t|c) = \phi(x_t, \varepsilon(c)) \tag{3}$$

This setup allows for iteratively refining an image c's estimated location from $t = 1$ to $t = 0$ via the Diffusion Denoising Implicit Models (DDIM) sampling procedure:

$$\hat{x}_t = \frac{1}{\sqrt{1-\kappa(t)}}(x_t - \sqrt{\kappa(t)}\psi(x_t|c) \tag{4}$$

Attack setup. In this work, following the literature [5], attacking the model means crafting a pixel-space adversarial perturbation δ^* to the source image which maximizes $\mathcal{D}(\hat{x}_0^{c+\delta^*}, x_0^c)$ the distance between the predicted and the real geographic coordinates of c. The distance measure $\mathcal{D}$ can vary, we use haversine distance. This perturbation δ^* is restricted to the b-ball $B_\infty(0, b)$, for b the budget, so as to control attack stealth. This constraint can be ensured by training the perturbation via projected gradient descent [15].

Attack costs. Training δ^* can be very expensive. As observed by Salman et al. [18], since accessing $\hat{x_0}^{c+\delta^*}$ consists in a full inference pass, the training process would require backpropagating through the whole diffusion process. As training memory costs are proportional to the number of inference steps, this makes direct training prohibitively expensive. We hence consider alternative training objectives, which are however all designed to craft an approximation of δ^*.

Encoder attack. A simple yet effective attack is to only focus on the encoder ε, rather than the diffusion process itself. This is very easy to implement because it only requires knowing the encoder, not the entire model. This attack is hence generalizable, and can even be used to attack multiple different models using the same encoder.

Salman et al. [18] proposed encoder attacks to shift image representations towards a latent target in the case of latent image diffusion. We mainly consider untargeted attacks, although our framework can be extended to targeted ones. Hence, we adapt Salman et al. [18]'s encoder attack to the following optimization problem:

$$\delta^{enc} = \underset{\|\delta\|_\infty \leq b}{argmax} ||\varepsilon(c+\delta) - \varepsilon(c)||^2 \tag{5}$$

We train a perturbation to shift the latent representation of the perturbed image to be as far away as possible from the source image under budget b.

Such attacks have been shown to be effective in the geo-privacy literature as mentioned in section **??**. As mentioned by Liu et al. [13], they however come at the expense of distorting all image semantics, as they are not targeted at geo-revealing features.

Diffusion Trajectory Deviation (DTD). Although the encoder attack is simple and generalizable, it may be limited as it only attacks the latent representation of the image, not directly its geo-revealing features. DTD mitigates this issue while still being cheap to train. It aims at crafting a perturbation which produces orthogonal diffusion trajectories. This can be achieved via the following training objective, which focuses on the forward diffusion process:

$$\delta^{DTD} = \underset{\|\delta\|_\infty \leq b}{\arg\min} \mathcal{L}^{DTD}(\psi, \delta), \tag{6}$$

$$\mathcal{L}^{DTD}(\psi, \delta) = \mathbb{E}_{x_0, c, \epsilon, t}\left[|\psi(x_t \mid c+\delta) \cdot \epsilon|\right] \tag{7}$$

δ^{DTD} can be trained just as one trains a diffusion model, via stochastic gradient descent on $\mathcal{L}^{DTD}$. This only requires backpropagating through ψ once, which is still more expensive than the encoder attack but cheaper than a direct attack by a factor of the number of inference steps.

3. Experiments

In this section, we first demonstrate the vulnerability of GPS metadata manipulation to diffusion geolocation (Sec. 3.1). We then evaluate our attacks in terms of average displacement caused and success rates (Sec. 3.2), explore the determinants of attack success (Sec. 3.3), and discuss our results and possible limitations of our attacks (Sec. 3.4).We base our work on two datasets:

- **OpenStreetView-5M** [1] **(OSV-5M)** with a ViT-L model [3] finetuned on Streetclip [7] as image encoder. We use a subset of the test set, composed of 200K images from 225 countries worldwide.
- **Yahoo Flickr Creative Commons 4K** [20] **(YFCC4k)**, a public subset of YFCC [20] comprised of 4000 generic images. We use it with DinoV2 [17] as an encoder.

These choices correspond to two of the three configurations used by Dufour et al. [4]. We also use their diffusion backbones and trained parameters for the aforementioned models. We only evaluate on small test subsets of YFCC and OSV-5M (100 images) due to constrained time and computing resources.

GPS metadata manipulation detection metrics. To evaluate our results regarding GPS metadata manipulation detection , we use the following metrics.

- **F1-score**. This score lies in $[0, 1]$. We use $\frac{2}{3}$ as a baseline, corresponding to the case where all samples are predicted as "true" (recall = 1 and precision = 0.5).
- **Accuracy**. This is a coefficient in $[0, 1]$. We use $\frac{1}{2}$ as a baseline, corresponding to naive strategies (always predicting true, always predicting false, or random guessing).

- **Matthews Correlation Coefficient (MCC)**. This coefficient lies in $[-1, 1]$. We use 0 as a baseline for the same reasons.

Attack metrics. We evaluate the effect of attacks via **Final Step Displacement** (FSD), that is the haversine distance in kilometers (km) between the predicted coordinates of the source and the perturbed image. Source coordinates are calculated using a classifier free guidance scale of 10 so as to focus the attack on the modes of the predicted coordinate distribution.

We also report attack **success rates**, that is the rate of images on which an attack has achieved a FSD greater than a given threshold. This allows to capture an attack's regularity, rather than its pure strength.

Attack success thresholds are set to 500, 1000, 1500 and 2500 km. The maximum threshold has been set following the literature on geolocation [4, 8, 13] where it is considered as the *continent level* threshold for computing geolocation success rates. We consider that distorting the predicted coordinates above this level bears little interest in terms of confidentiality.

Attack hyperparameters. Throughout our experiments, attacks are trained on each image over 80 steps, a batch-size of 256 (when applicable, i.e. for DTD) and a learning rate of 10^{-3}. Attacks are trained with 6 independent starting points, and we select the one achieving the highest FSD in order to improve attack consistency.

3.1. Manipulating GPS metadata does not protect privacy

We first try to evaluate how current models are able to distinguish fakes locations using the two aforementioned methods (threshold-based method and localizability-based method).

Crafting false geolocations. The experiments run with an equal number of real and fake geolocations. For generating false locations, we rely on two different methods:

- **Shuffle method**. We associate to an image the geolocation of another image. This ensures that the fake-geolocation is a real location from which the image can actually come.
- **Distance method**. We associate to an image a random geolocation at a distance d from the real one. In our experiments, $d = 100km$ or $d = 1000km$

Results. (See Figure 5 to 8 for more detailed results.)

Both methods (threshold-based and localisability-based) show clear potential. We achieve modest but consistent results for most benchmarks (notably with an accuracy > 0.6 and a MCC > 0.2).

The OSV dataset with negatives generated 100km away of the true location is however not handled properly by either of the methods. This probably lies in the very nature of the images that do not contain clear signs enabling such a precision.

Both methods give roughly similar results when using the best settings. However, the use of the parameter $\Delta = -4$ gives close to the best results in any situation whereas for using the first method optimally, we have to adapt the threshold depending on the dataset (0 for YFCC and -1.5 for OSV)

3.2. Performance of encoder and diffusion attacks

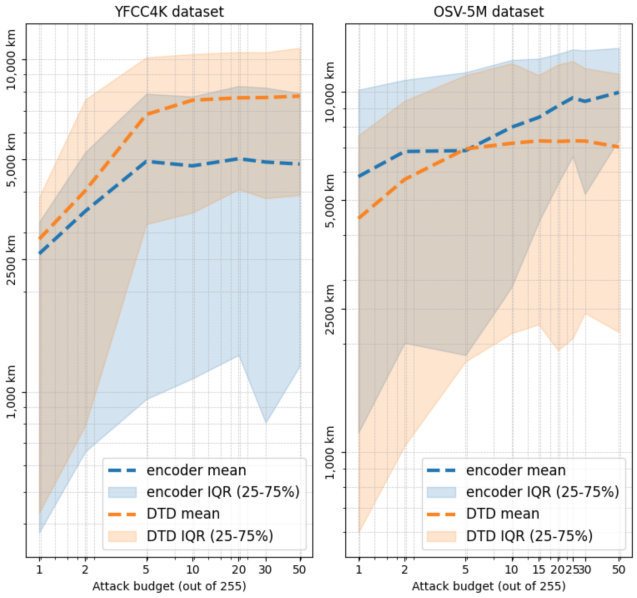

Figure 1. Final step displacement of encoder and DTD attacks on 100 random images of YFCC4k and OSV5M (test set), according to attack budget. Interquartile ranges are reported for each attack. All scales are logarithmic.

We then turn to evaluating comparatively our encoder and DTD attacks as means to protect the geoprivacy of images. Figure 1 displays average FSD of encoder and DTD attacks for the OSV and YFCC4k datasets, while figures 2 and 3 report their success rates over several success thresholds.

Attack stealth. We observe that both attacks can be efficient at very low attack budgets. For budgets below the range of detectability to the human eye, $\frac{2}{255}$, average FSDs are respectively of the order of 5000 and 2500 km for OSV-5M and YFCC4k respectively. While both FSD and success rates improve with attack budget, most of the improvement occurs between $\frac{1}{255}$ and $\frac{10}{255}$. At $\frac{10}{255}$, the best attack of the two reaches a success rate over 75% for the 2500 km threshold. These values are lower than the budget considered by Salman et al. [18] ($\frac{16}{255}$), and comparable to state of the art stealth utilized in the literature [2, 11] ($\frac{4}{255}$ to $\frac{8}{255}$), and in the geo-privacy protection literature ($\frac{8}{255}$ to $\frac{16}{255}$) [13].

Attack reliability. We observe that FSD exhibits high

variance for both attacks despite selecting the best out of 6 training seeds. The interquartile range spans over 4000 to 9000 km dpeending on the attack and dataset for low budgets, and slightly reducing for higher budgets. This indicates that attack dynamics (and success) are highly image dependent, and calls for further investigation of the determinants of attack success.

Comparison of encoder and DTD attacks. Surprisingly, the encoder and DTD attacks exhibit opposite relative results on OSV-5M and YFCC4k. The encoder attack slightly surpasses DTD on OVS-5M in most cases, that is when it attacks the ViT encoder fine-tuned of *StreetClip*. We note DTD however has a higher 2500km threshold success rate at budget $\frac{5}{255}$. It is also more successful than the encoder attack on images from YFCC4K (and hence the DinoV2-backbone) in terms of success rate, FSD and interquartile range above budget $\frac{5}{255}$. Overall, attacks on OSV-5M images, that is on the model using the *StreetClip* DiT, achieve higher FSDs over all attack budgets. We however observe that at stealthy budgets ($\frac{2}{255}$ to $\frac{10}{255}$), at least one of the attacks achieves a success rate of around 80%.

General performance Average FSD of DTD is comparable in magnitude to results against VLMs reported (5000 to 8000 km) for *GeoShield* [13] at comparable budgets ($\frac{10}{255}$ to $\frac{15}{255}$).

3.3. Determinants of attack success

In this section, we explore the determinants of attack success. Figure 4 displays average FSD distributions for low, intermediate and highly localizable images. We observe a strong positive correlation between localizability and FSD on YFCC4K. Results on OSV-5M are more difficult to interpretate, although highly localizable images remain the ones that are the easiest to attack.

The encoder attack seems especially ineffective against low localizable images of YFCC4K (FSD under 1000 km), while it maintains high FSD values for medium and high localizability. DTD also loses strength when localizability is low, but still succesfully distorts the model at these levels with an average DTD of 5000 km. Finally, we observe that the more localizable an attack is, the more reliable it is, i.e. the lower DTD variance is.

3.4. Discussion

Our results demonstrate the effectiveness and efficiency of stealthy DTD attacks across datasets, with a 2500 km-success rate above 75% for a budget of $\frac{10}{255}$ over both datasets. It hence seems to be an effective method to protect the geo-privacy of images. The performance of encoder attacks is more inconsistent across datasets. It is very efficient against the DiT encoder, even more than DTD, but displays persistent failures against DinoV2 with a 2500 km success rate rate under 62% for all budgets. This is likely due to increased robustness of DinoV2.

Whether this is thanks to its unsupervised training agenda targetting blur and noise invariance, or to its higher latent dimensionality, is beyond the scope of this paper. Critically, we notice that encoder attacks perform badly precisely when the geo-privacy preservation task seems the most difficult - i.e. the datasets for which both attacks display the lowest FSD values. In the contrary, DTD displays consistent performance accross dataset-model pairs.

The observed positive correlation between attacks success, both in terms of strength and reliability, and localizability could imply that preserving privacy is the easiest when a precise position can be attributed to an image. In the contrary, attack failures happen when the predicted positions of an image are widespread. This indicates that our attacks are the best when they are the most needed, confirming their usefulness to protect geo-privacy.

3.5. Limitations

Our framework and evaluation method presents several limitations. First of all, we consider white box attacks, i.e. attacks where we know the model architecture and its weights. In the contrary, Liu et al. [13] propose a black box attack which is transferable accross VLMs. In that sense, our attacks remain limited in scope. Moreover, we only evaluated our framework on 100 images of each dataset due to time and compute constraints. This might explain the high variability in FSD observed, and limits the scope of our results. Moreover, we have used the datasets on which Dufour et al. [4] trained their diffusion models, limiting comparability with the literature.

Finally, we have focused our adversarial framework (encoder and DTD attacks) on geographic diffusion, rather than its flow matching and Riemmanian flow matching extensions [4]. The intuition backing DTD remains however applicable for these extensions. We leave its application to future work.

4. Conclusion

In this work, we have introduced the task of preventing visual geolocation of images via conditional geographic diffusion, after having outlined the effectiveness of this method to detect GPS metadata manipulation. Our geo-privacy preserving methods rely on crafting adversarial perturbations to an image. We considered two attacks, one existing encoder attack, and our novel Diffusion Trajectory Deviation attack which directly targets the geographic diffusion process.

Our attacks display high success rates for state-of-the-art attack budgets, effectively preventing images from being localized. Our DTD attack displays particular consistency across encoders and datasets, paving the way for geo-privacy specific attacks.

5. Appendix

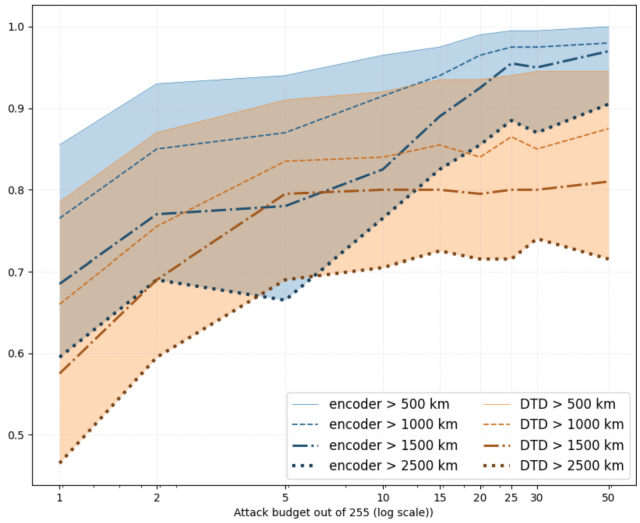

Figure 2. Success rates of encoder and DTD attacks on 100 random images of OSV-5M's test set, according to log-attack budget, for several success thresholds.

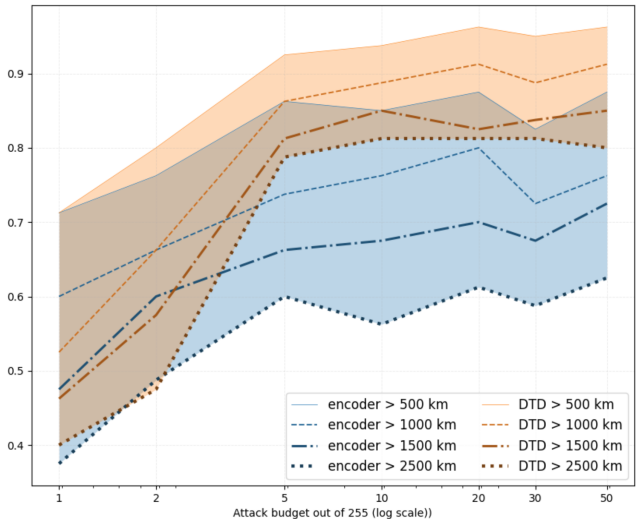

Figure 3. Success rates of encoder and DTD attacks on 100 random images of YFCC4k, according to log-attack budget, for several success thresholds.

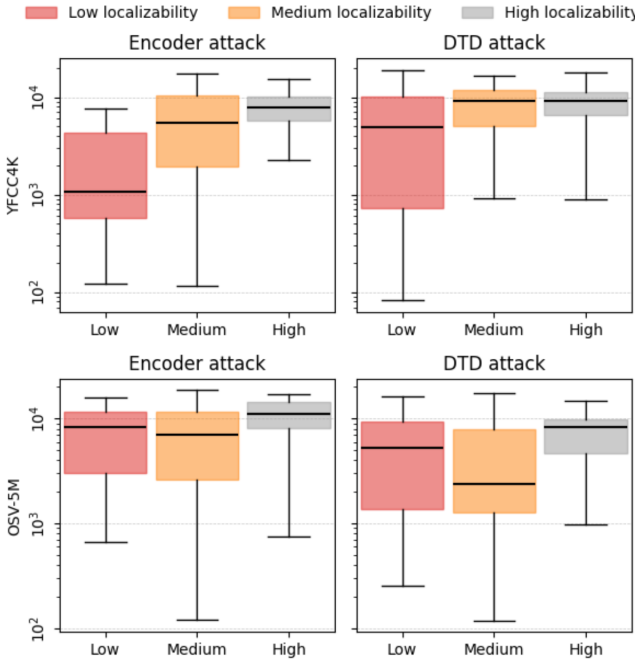

Figure 4. Distributions of final step displacement of encoder and DTD attacks on YFCC4K and OSV-5M according to localizability. Low, medium and high localizability thresholds are designed using 33 and 66% quantiles of localizabiity.

THRESHOLD - METHOD	Negatives	YFCC		OSV	
		Score	Threshold	Score	Threshold
F1-Score	Shuffle	**0.69**	-1.6	0.76	-2.0
	100 km	**0.67**	-3.3	**0.67**	-9.5
	1000 km	**0.76**	-1.1	**0.67**	-2.4
Accuracy	Shuffle	0.63	-0.1	**0.75**	-1.6
	100 km	0.61	0.9	**0.52**	0.5
	1000 km	**0.75**	-0.1	0.61	-1.2
MCC	Shuffle	0.27	-0.1	**0.51**	-1.5
	100 km	0.24	1.6	**0.05**	0.5
	1000 km	**0.50**	-0.1	0.24	-0.1

Figure 5. Scores of the best thresholds for the threshold-method

LIKELIHOOD - METHOD	Negatives	YFCC		OSV	
		Score	Delta	Score	Delta
F1-Score	Shuffle	**0.69**	-6.1	0.73	-6.4
	100 km	**0.68**	-7.0	**0.67**	-11.8
	1000 km	**0.76**	-3.8	**0.69**	-6.9
Accuracy	Shuffle	0.64	-3.8	**0.70**	-4.2
	100 km	0.62	-2.9	**0.53**	-6.0
	1000 km	**0.75**	-3.8	0.61	-4.1
MCC	Shuffle	0.29	-3.8	**0.40**	-4.1
	100 km	0.24	-2.9	**0.08**	-6.0
	1000 km	**0.50**	-3.8	0.23	-4.1

Figure 6. Scores of the best deltas for the likelihood-method

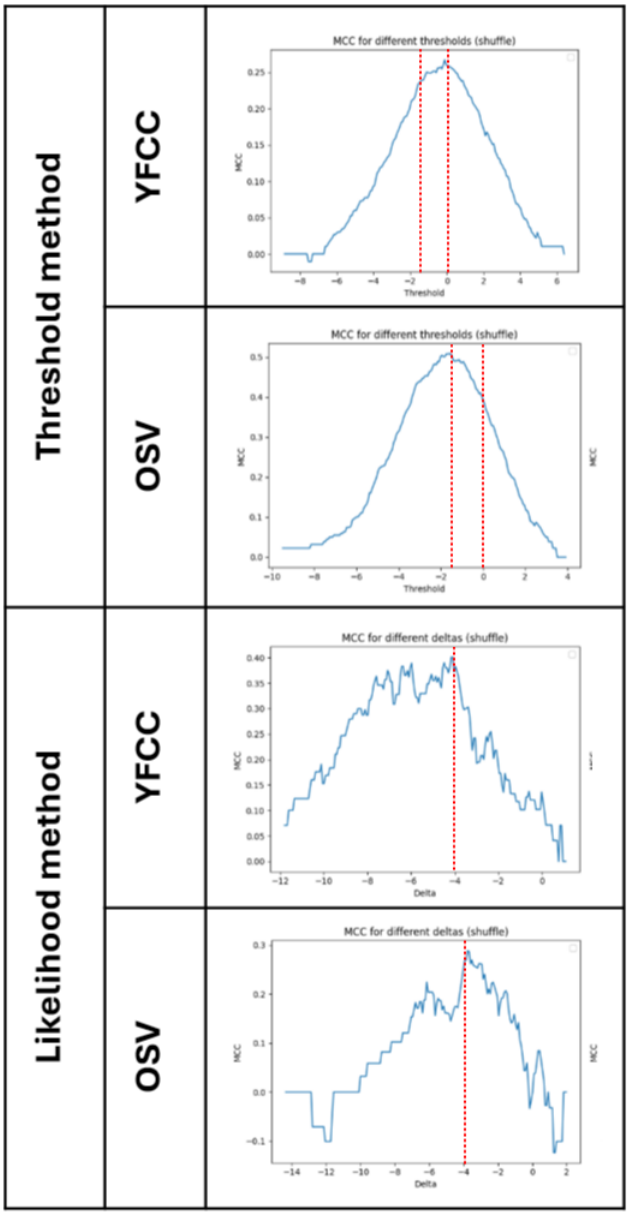

Figure 7. Comparisons of the MCC when the negatives are shuffled

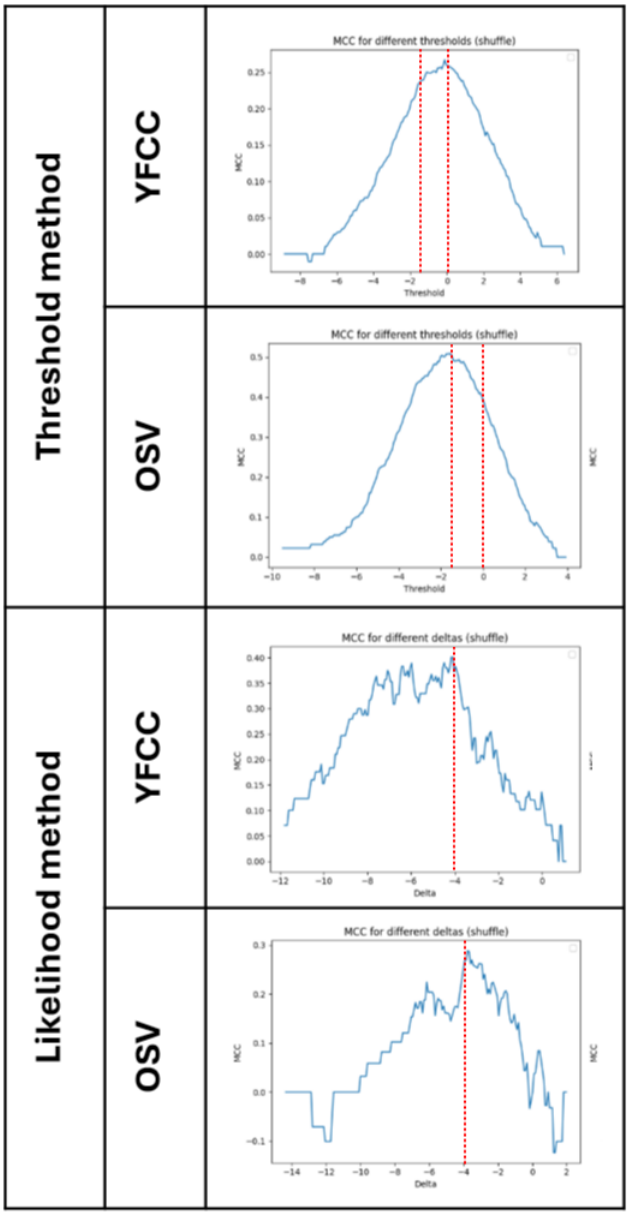

Figure 8. Comparisons of the MCC when the negatives are taken randomly at 1000km

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